

Cooperative UAV Jammer for Enhancing Physical Layer Security: Robust Design for Jamming Power and Trajectory

Yujung Roh, Seungjae Jung, and Joonhyuk Kang
School of Electrical Engineering

Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Korea
E-mail : {c15972, sjjung513}@kaist.ac.kr, and jhkang@ee.kaist.ac.kr

Abstract—An unmanned aerial vehicle (UAV)-aided network is becoming a promising application for the future wireless communication due to the flexible deployment and dominant line-of-sight channel. In this paper, we consider the UAV is operated as a cooperative jammer to enhance the physical layer security of the ground legitimated nodes in the presence of an eavesdropper (Eve). Furthermore, we assume that the UAV has imperfect information on the locations of the receiver and Eve due to GPS jamming and covert operation of Eve, respectively. With these uncertainties of the nodes' locations, we formulate a robust joint optimization problem of the UAV's jamming power and trajectory to maximize the average secrecy rate. To handle the non-convexity of the optimization problem, we propose an iterative suboptimal algorithm based on the block coordinate descent method. Simulation results present that the proposed algorithm has outstanding performance in terms of physical layer security compared to other benchmark methods.

Index Terms—UAV-aided communications, physical layer security, cooperative UAV jammer, power allocation, trajectory design

I. INTRODUCTION

As an unmanned aerial vehicle (UAV) technology has been developed, it is actively exploited in a variety of wireless communication system from a cellular network to a battlefield [1]–[11]. Due to high mobility and flexible deployment, the UAV can achieve a relatively dominant line-of-sight (LoS) channel between the ground node (GN) in comparison with the ground communication. Motivated by this advantage, secure communication with the UAV jammer has been currently investigated in [7]–[9]. A conventional ground jammer generally has the apparent limitation: the static location and the fading channel. On the other hand, the UAV jammer can improve the physical layer secrecy of GNs by designing the proper trajectory and transmitting the artificial noise to the eavesdropper (Eve) via LoS link.

Generally, the UAV is operated as a mobile transmitter or a receiver for communicating with ground nodes for secure communication in [10], [11], while it is employed as a cooperative jammer to disturb Eve in [8], [9]. These previous research jointly optimize the trajectory and the transmitting power of the UAV to maximize the secrecy rate of the GNs, with exact information on Eve's location. Meanwhile, recently, a few of studies are carried out in consideration of more

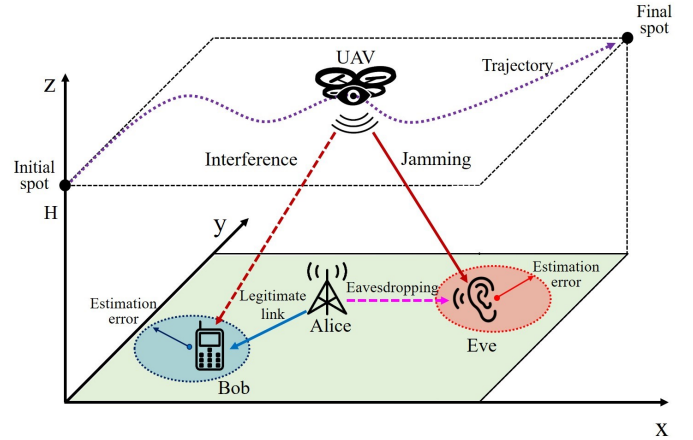


Fig. 1. A ground wiretap communication with a cooperative UAV jammer

realistic situation where the UAV has imperfect information on the location of Eve in [5]–[7]. In [5], [6], the uncertainty of Eve's location is addressed for trajectory design and power allocation of the UAV transmitter. In [7], the dual UAV system as a transmitter and a jammer is investigated with the imperfect location of Eve. On the other hand, in the more practical situation such as the battlefield, it should be assumed that not only Eve's location but also the locations of the legitimated nodes are uncertain due to the enemy's disturbance activity such as GPS jamming [12], [13].

In this paper, a cooperative UAV jammer scenario to improve the secrecy rate of the legitimated GNs is considered. Motivated by the limitation of previous studies, we assume that the UAV jammer partially recognizes the locations of the receiver and Eve with the estimation errors. It is assumed that the uncertainty of the legitimated receiver is caused by GPS jamming. Furthermore, the UAV jammer can gather information on the estimated location of Eve by using the UAV's surveillance assets such as a camera or spectrum tracking. We formulate the joint optimization problem of the UAV jamming power and the trajectory for enhancing the secrecy rate. To make this problem more tractable, we derive the secrecy rate of the worst case regarding to the nodes' estimated locations. Since our optimization problem turns out to be non-convex, we

divide the problem into two sub-problems and propose a robust iterative suboptimal algorithm based on the block coordinate descent (BCD) method. In addition, our study investigates the UAV trajectory and secrecy rate depending on the estimation errors of the nodes' locations. Via simulation results, we can observe the tendency of the optimal UAV trajectory and the achievable secrecy rate to change with the estimation errors and the UAV's mission time.

This paper is structured as follows. Section II defines the system model of the problem. Section III describes formulations of optimization problem and an efficient iterative algorithm. The simulation results are presented in Section IV and conclusions are drawn in Section V.

II. SYSTEM MODEL

We consider a cooperative UAV jammer for secure ground communication, which consists of a transmitter (Alice), a legitimated receiver (Bob), and Eve, as shown in Fig. 1. It is considered that Alice, Bob and Eve are located at $\mathbf{L}_i = [x_i, y_i]^T, i \in \{A, B, E\}$ in meter (m) on the ground, respectively. In addition, it is assumed that the UAV perfectly knows Alice's location by cooperative communication. On the other hand, the UAV has imperfect information on locations of Bob and Eve due to the enemy's possible disturbance (e.g. GPS jamming and covert operation of Eve). Hence, we consider norm-bounded error model in [6] for representing the uncertainties of Bob and Eve's location. Let $\tilde{\mathbf{L}}_i = [\tilde{x}_i, \tilde{y}_i]^T, i \in \{B, E\}$ denote the estimated node's location. Thus, the exact location of the node is given by

$$\mathbf{L}_i = \tilde{\mathbf{L}}_i + \Delta\mathbf{L}_i, i \in \{B, E\}, \quad (1)$$

where $\Delta\mathbf{L}_i = [\Delta x_i, \Delta y_i]^T$ denotes the estimation error. It satisfies the following condition by norm-bounded error model:

$$\|\Delta\mathbf{L}_i\|^2 \leq \epsilon_i^2, i \in \{B, E\}, \quad (2)$$

where ϵ_i denotes the maximum magnitude of the estimation error. It means that the node can be located in an uncertain circular region with a radius ϵ_i around the estimated location $\tilde{\mathbf{L}}_i$.

We assume that the UAV jammer is operated from predetermined location from $\mathbf{u}_I$ to $\mathbf{u}_F$ at constant altitude H to support a certain military mission during flight period T . To obtain a more tractable problem, we consider that the mission time T is discretized into N time slots with a sufficiently small time $t_s = T/N$ in second (s). In addition, $d_s = Vt_s$ denotes the maximum distance of horizontal traveling within two consecutive time slots with maximum speed V in m/s, and N must be chosen in order to d_s is small enough for H so that no change in the air-to-ground channels of the UAV occurs within each slot. Let $\mathbf{u}[n] = [x[n], y[n]]^T$ denote the horizontal location of the UAV in time slot $n \in \{1, \dots, N\}$, then it satisfies mobility constraints as follows:

$$\|\mathbf{u}[n+1] - \mathbf{u}[n]\|^2 \leq d_s^2, n = 1, \dots, N-1, \quad (3)$$

$$\|\mathbf{u}[1] - \mathbf{u}_I\|^2 \leq d_s^2, \quad (4)$$

$$\mathbf{u}[N] = \mathbf{u}_F. \quad (5)$$

The channels between the UAV jammer and GNs are modeled as free-space wireless channel due to the dominant LoS link. Thus, the channel coefficient from the UAV to the GN i in time slot n is given by

$$h_{Ui}[n] = \rho_0 d_{Ui}^{-2}[n] = \frac{\rho_0}{\|\mathbf{u}[n] - \mathbf{L}_i\|^2 + H^2}, i \in \{B, E\}, \quad (6)$$

where $d_{Ui}[n]$ is the distance between the UAV and the GN i in time slot n , and ρ_0 denotes the gain of a channel with the reference distance $d_0 = 1$ m. On the other hand, the channels from Alice to Bob and Eve are assumed to be Rayleigh channel. Thus, the channel coefficient from Alice in slot n is given by

$$g_{Ai} = \rho_0 d_{Ai}^{-\psi} \zeta_i, i \in \{B, E\}, \quad (7)$$

where ψ denotes the path loss exponent; ζ_i is the Rayleigh channel coefficient with complex Gaussian with zero mean and unit variance. The channels are stationary and ergodic within two consecutive time slots, since it is assumed that t_s is generally much smaller than the coherence time of ground channels.

Moreover, the transmit power of the UAV jammer, $P_U[n]$, in time slot n should be satisfied with average and peak power in the following constraints:

$$\frac{1}{N} \sum_{n=1}^N P_U[n] \leq P_U^{\text{avg}}, \quad (8)$$

$$0 \leq P_U[n] \leq P_U^{\text{max}}, \quad (9)$$

where P_U^{avg} and P_U^{max} are denoted as average and maximum transmit power of the UAV jammer, respectively. Then, the achievable rates from Alice to Bob and Eve in bits/s/Hz (bps/Hz) in time slot n are expressed as

$$R_{AB}[n] = \mathbb{E} \left[\log_2 \left(1 + \frac{g_{AB} P_A}{h_{UB}[n] P_U[n] + \sigma^2} \right) \right], \quad (10)$$

$$R_{AE}[n] = \mathbb{E} \left[\log_2 \left(1 + \frac{g_{AE} P_A}{h_{UE}[n] P_U[n] + \sigma^2} \right) \right], \quad (11)$$

where $\mathbb{E}[\cdot]$ is the expectation with respect to independent Rayleigh fading channel; σ^2 is the additive white Gaussian noise at the GNs; P_A is the transmit power of Alice. Therefore, the secrecy rate in time slot n can be formulated by

$$[R[n]]^+ = [R_{AB}[n] - R_{AE}[n]]^+, \quad (12)$$

where $[x]^+ \triangleq \max(x, 0)$ means that the secrecy rate in each time slot n must be non-negative.

III. PROBLEM FORMULATION AND ALGORITHM PROPOSAL

In this paper, we aim to maximize the average secrecy rate of legitimated GNs for total mission period. To achieve this goal, we focus on the joint optimization of the jamming power and the flight trajectory of the UAV jammer. Thus, the optimization problem of the average secrecy rate can be expressed as

$$\begin{aligned} \text{P1 : } & \max_{P_U[n], \mathbf{u}[n]} \frac{1}{N} \sum_{n=1}^N R[n] \\ & \text{s.t. } (3), (4), (5), (8), (9), \end{aligned} \quad (13)$$

where the operator $[\cdot]^+$ in (12) is omitted to tackle the non-smoothness of the problem by using Lemma 1 in [5].

For guaranteeing the physical layer secrecy of the legitimated GNs, we consider the achievable rate of the worst case according to [7] as follows:

$$\tilde{R}_{Ai}[n] = \mathbb{E} \left[\log_2 \left(1 + \frac{\tilde{g}_{Ai} P_A}{\tilde{h}_{Ui}[n] P_U[n] + \sigma^2} \right) \right], \quad (14)$$

where $\tilde{h}_{Ui}[n]$, $i \in \{B, E\}$ denotes the worst channel coefficient between the UAV and the node; $\tilde{g}_{Ai}$ denotes the worst channel coefficient from Alice to the node. The worst channel coefficients with Bob's uncertainty can be obtained by

$$\max_{\mathbf{L}_B} h_{UB}[n] = \tilde{h}_{UB}[n] = \frac{\rho_0}{(|\mathbf{u}[n] - \tilde{\mathbf{L}}_B| - \epsilon_B)^2 + H^2}, \quad (15)$$

$$\min_{\mathbf{L}_B} g_{AB} = \tilde{g}_{AB} = \rho_0 (\tilde{d}_{AB} + \epsilon_B)^{-\psi} \zeta_B. \quad (16)$$

Similarly, the worst channel coefficients considering Eve's uncertainty can be obtained by

$$\min_{\mathbf{L}_E} h_{UE}[n] = \tilde{h}_{UE}[n] = \frac{\rho_0}{(|\mathbf{u}[n] - \tilde{\mathbf{L}}_E| + \epsilon_E)^2 + H^2}, \quad (17)$$

$$\max_{\mathbf{L}_E} g_{AE} = \tilde{g}_{AE} = \rho_0 (\tilde{d}_{AE} - \epsilon_E)^{-\psi} \zeta_E. \quad (18)$$

We define the secrecy rate of the worst case as $\tilde{R}[n] = \tilde{R}_{AB}[n] - \tilde{R}_{AE}[n]$. Thus, P1 can be formulated as

$$\begin{aligned} \text{P2 : } \max_{P_U[n], \mathbf{u}[n]} & \quad \frac{1}{N} \sum_{n=1}^N \tilde{R}[n] \\ \text{s.t. } & \quad (3), (4), (5), (8), (9). \end{aligned} \quad (19)$$

Since the achievable rates of P2 include the expectation of channel fading, it is difficult to find the optimal solution. To tackle complexity of this problem, we consider the approximate bounds of the achievable rates for Bob and Eve based on [9], [14]. First, according to the convexity of $\ln(1 + e^x)$ and Jensen's inequality, the lower bound $\tilde{R}_{AB}^{\text{lo}}[n]$ of $\tilde{R}_{AB}[n]$ can be obtained as

$$\tilde{R}_{AB}[n] \geq \tilde{R}_{AB}^{\text{lo}}[n] = \log_2 \left(1 + \frac{\hat{g}_{AB} P_A}{\hat{h}_{UB}[n] P_U[n] + 1} \right), \quad (20)$$

where $\hat{g}_{AB} = e^{-\gamma} \alpha_0 (\tilde{d}_{AB} + \epsilon_B)^{-\psi}$ and $\hat{h}_{UB}[n] = \frac{\alpha_0}{(|\mathbf{u}[n] - \tilde{\mathbf{L}}_B| - \epsilon_B)^2 + H^2}$. In addition, $\alpha_0 = \rho/\sigma^2$ and γ is the Euler constant. Similarly, using the concavity of $\ln(1 + x)$, the upper bound $\tilde{R}_{AE}^{\text{up}}[n]$ of $\tilde{R}_{AE}[n]$ is given by

$$\tilde{R}_{AE}[n] \leq \tilde{R}_{AE}^{\text{up}}[n] = \log_2 \left(1 + \frac{\hat{g}_{AE} P_A}{\hat{h}_{UE}[n] P_U[n] + 1} \right), \quad (21)$$

where $\hat{g}_{AE} = \alpha_0 (\tilde{d}_{AE} - \epsilon_E)^{-\psi}$, $\hat{h}_{UE}[n] = \frac{\alpha_0}{(|\mathbf{u}[n] - \tilde{\mathbf{L}}_E| + \epsilon_E)^2 + H^2}$. Hence, we define the approximate secrecy rate as $\hat{R}[n] = \tilde{R}_{AB}^{\text{lo}}[n] - \tilde{R}_{AE}^{\text{up}}[n]$, and P2 can be rewritten as

$$\begin{aligned} \text{P3 : } \max_{P_U[n], \mathbf{u}[n]} & \quad \frac{1}{N} \sum_{n=1}^N \hat{R}[n] \\ \text{s.t. } & \quad (3), (4), (5), (8), (9). \end{aligned} \quad (22)$$

However, P3 is a still non-convex problem due to difference of convex (DC) function with respect to $P_U[n]$ and $\mathbf{u}[n]$. To handle this problem, we propose an iterative algorithm using the BCD method. Hence, P3 is divided into two sub-problems to optimize $P_U[n]$ and $\mathbf{u}[n]$ alternately.

A. Robust power allocation of UAV jamming

In this step, we formulate the optimization problem depending on $P_U[n]$ while assuming that the other variable $\mathbf{u}[n]$ is given. Applying the first-order Taylor expansion, we obtain the lower bound $\hat{R}^{(k)}[n]$ of $\hat{R}[n]$ is given by

$$\begin{aligned} \hat{R}[n] & \geq \hat{R}^{(k)}[n] \\ & = \log_2 \left(1 + \frac{\hat{g}_{AB} P_A}{\hat{h}_{UB}[n] P_U^{(k)}[n] + 1} \right) \\ & \quad - a_n^{(k)} (P_U[n] - P_U^{(k)}[n]) \\ & \quad - \log_2 \left(1 + \frac{\hat{g}_{AE} P_A}{\hat{h}_{UE}[n] P_U[n] + 1} \right), \end{aligned} \quad (23)$$

where $a_n^{(k)} = \frac{\hat{g}_{AB} P_A \hat{h}_{UB}[n]}{\ln 2 (\hat{h}_{UB}[n] P_U^{(k)}[n] + 1) (\hat{h}_{UB}[n] P_U^{(k)}[n] + \hat{g}_{AB} P_A + 1)}$. Also $P_U^{(k)}[n]$ denotes the feasible value of k -th iteration about $P_U[n]$ on the first term of P3. Then P3 can be written as

$$\begin{aligned} \text{P4 : } \max_{P_U[n]} & \quad \frac{1}{N} \sum_{n=1}^N \hat{R}^{(k)}[n] \\ \text{s.t. } & \quad (8), (9). \end{aligned} \quad (24)$$

Since P4 is a convex programming problem, we can obtain the optimal solution by interior point method in CVX tool.

B. Robust trajectory design for UAV jamming

In this subsection, we assume that $P_U[n]$ is given, and reformulate $\hat{R}[n]$ of the P3 to $\check{R}[n]$ for $\mathbf{u}[n]$ by introducing slack variable $b[n]$ and $c[n]$.

$$\check{R}[n] = \log_2 \left(1 + \frac{\hat{g}_{AB} P_A}{\frac{\alpha_0 P_U[n]}{b[n]} + 1} \right) - \log_2 \left(1 + \frac{\hat{g}_{AE} P_A}{\frac{\alpha_0 P_U[n]}{c[n]} + 1} \right), \quad (25)$$

where constraints of slack variables are given by

$$b[n] \leq (|\mathbf{u}[n] - \tilde{\mathbf{L}}_B| - \epsilon_B)^2 + H^2, \quad (26)$$

$$c[n] \geq (|\mathbf{u}[n] - \tilde{\mathbf{L}}_E| + \epsilon_E)^2 + H^2. \quad (27)$$

Equation (25) is also non-convex due to DC function depending on $\mathbf{u}[n]$. In the same manner as the approximation of $\hat{R}[n]$, by applying the first-order Taylor expansion, the lower bound $\check{R}^{(k)}[n]$ of $\check{R}[n]$ is reformulated by

$$\begin{aligned} \check{R}[n] & \geq \check{R}^{(k)}[n] \\ & = \log_2 \left(1 + \frac{\hat{g}_{AB} P_A}{\frac{\alpha_0 P_U[n]}{b[n]} + 1} \right) - \log_2 \left(1 + \frac{\hat{g}_{AE} P_A}{\frac{\alpha_0 P_U[n]}{c^{(k)}[n]} + 1} \right) \\ & \quad - e_n^{(k)} (c[n] - c^{(k)}[n]), \end{aligned} \quad (28)$$

where $e_n^{(k)} = \frac{\hat{g}_{AE} P_A \alpha_0 P_U[n]}{\ln 2 (c^{(k)}[n] + \alpha_0 P_U[n]) ((\hat{g}_{AE} P_A + 1) c^{(k)}[n] + \alpha_0 P_U[n])}$; $c^{(k)}[n]$ is the feasible value of k -th iteration about $c[n]$ on

Algorithm 1 Iterative robust algorithm via joint optimization of the jamming power and the trajectory

- 1: Initialize $P_U^{(0)}[n]$ and $\mathbf{u}^{(0)}[n]$. Set $k = 0$.
- 2: **repeat**
- 3: Given $P_U^{(k)}[n]$ and $\mathbf{u}^{(k)}[n]$, find the optimal solution $P_U^{(k+1)}[n]$ of problem P4.
- 4: Given $P_U^{(k+1)}[n]$ and $\mathbf{u}^{(k)}[n]$, find the optimal solution $\mathbf{u}^{(k+1)}[n]$ of problem P5.
- 5: Update $k \leftarrow k + 1$.
- 6: **until** convergence.

TABLE I
SIMULATION PARAMETER

Parameter	Value
Altitude of the UAV, H	100 (m)
Maximum speed of the UAV, V	3 (m/s)
Number of time slots, N	50
Path loss exponent, ψ	3
Channel gain over noise power, $\alpha_0 = \rho_0/\sigma^2$	90 (dB)
Transmit power of Alice, P_A	30 (dBm)
Average power of the UAV, P_U^{avg}	10 (dBm)
Maximum power of the UAV, P_U^{max}	$4P_U^{\text{avg}}$

the second term of (25). Similarly, a constraint (26) can be rewritten as

$$b[n] \leq f_n^{(k)} + H^2, \quad (29)$$

where $f_n^{(k)} = (||\mathbf{u}^{(k)}[n] - \tilde{\mathbf{L}}_B|| - \epsilon_B)^2 + (2[\mathbf{u}^{(k)}[n] - \tilde{\mathbf{L}}_B]^T - 2\epsilon_B \frac{[\mathbf{u}^{(k)}[n] - \tilde{\mathbf{L}}_B]^T}{||\mathbf{u}^{(k)}[n] - \tilde{\mathbf{L}}_B||})(\mathbf{u}[n] - \mathbf{u}^{(k)}[n])$, and $\mathbf{u}[n]$ of (26) is set to the feasible value of k -th iteration $\mathbf{u}^{(k)}[n]$. Then, P3 can be represented as

$$\begin{aligned} \text{P5 : } & \max_{\mathbf{u}[n], b[n], c[n]} \frac{1}{N} \sum_{n=1}^N \tilde{R}^{(k)}[n] \\ & \text{s.t. } (3), (4), (5), (27), (29). \end{aligned} \quad (30)$$

Since P5 is also a convex programming problem, the optimal solution can be calculated by interior point method in CVX tool.

C. Iterative robust algorithm

In this subsection, we propose the iterative robust algorithm (RA) to jointly optimize the jamming power and the trajectory of the UAV jammer. As shown in Algorithm 1, RA executes the procedure for finding the suboptimal solution of P3 by repeatedly solving sub-problems P4 and P5 until the solution converges. The $(k+1)$ -th solution can be converged without decreasing over iteration via the BCD method and the non-convexity approximation of the subsections.

IV. SIMULATION RESULTS

In this section, we evaluate the performance of the proposed algorithm RA via simulation results. For verifying the enhancement of RA, we consider following two methods.

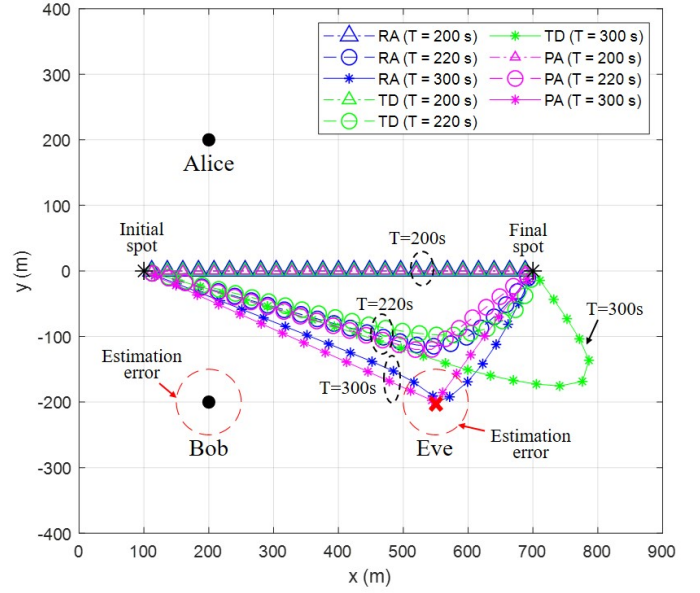


Fig. 2. UAV trajectory with respect to the mission time T , when Bob's estimation error $\epsilon_B = 50$ m and Eve's estimation error $\epsilon_E = 50$ m

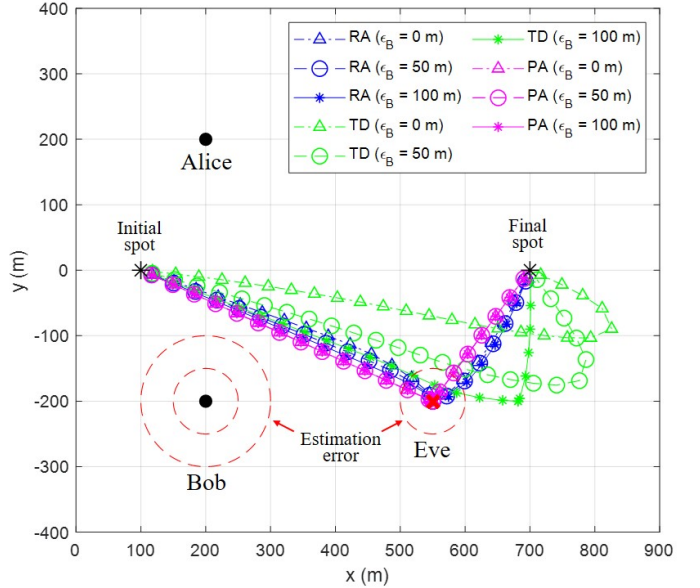


Fig. 3. UAV trajectory with respect to Bob's estimation error ϵ_B , when the mission time $T = 300$ s

- 1) Only trajectory design (TD): the trajectory is designed optimally in the same way as RA, but the UAV jamming power is fixed at P_U^{avg} .
- 2) Only power allocation (PA): the UAV jamming power is optimized by RA, but the trajectory design is given by following conditions:
 - Sufficient mission time case: the UAV flies straight from the initial spot to the above point of Eve's estimated location and hovers during the available time. Then, the UAV flies directly to the final spot at

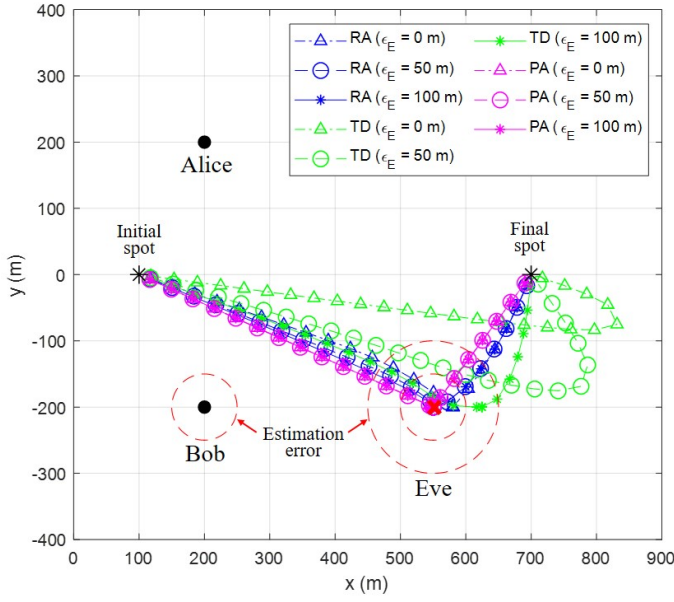


Fig. 4. UAV trajectory with respect to Eve's estimation error ϵ_E , when the mission time $T = 300$ s

the maximum speed during mission time remaining.

- Insufficient mission time case: the UAV flies as close as possible to the estimated location of Eve and then turns to the final spot at the maximum speed.

The parameters of the simulations are set to practical values as shown in Table I with reference to [7], [9], and the coordinates of GNs are as follows: $\mathbf{L}_A = [200, 200]^T$, $\mathbf{L}_B = [200, -200]^T$, $\mathbf{L}_E = [550, -200]^T$, $\mathbf{u}_I = [100, 0]^T$, and $\mathbf{u}_F = [700, 0]^T$.

Fig. 2 shows the trajectories over the mission time T according to RA, TD, and PA. In this figure, it is considered that the estimation errors ϵ_B and ϵ_E set 50 m. When $T = 200$ s, it is observed that there is no difference in the trajectories of the three methods, since T is the minimum time needed to travel from the initial spot to the final spot at the maximum speed. On the other hand, when the mission time is sufficient to reach Eve (i.e. $T = 300$ s), the UAV tries to approach Eve and move away as far as possible from Bob simultaneously. This is because the UAV jamming is also interference signal to the legitimated receiver, Bob. Therefore, the UAV tends to find the optimal trajectory where it can send strong jamming signal to Eve and low interference signal to Bob at the same time. Specially, in TD, the UAV tends to avoid Bob rather than approach Eve, since it cannot control the jamming power.

Fig. 3 and Fig. 4 show the trajectories with respect to the estimation errors ϵ_B and ϵ_E , respectively, given $T = 300$ s. As ϵ_B and ϵ_E increase, the trajectory of RA becomes similar to that of PA. Similarly, in TD, the UAV tends to approach Eve, when the estimation errors grow. It is because the UAV tries to get closer to Eve in order to attenuate the loss of secrecy rate due to the uncertainties of Bob and Eve.

Fig. 5 demonstrates the tendency of the average secrecy rate versus the mission time T . For comparison, we evaluate the

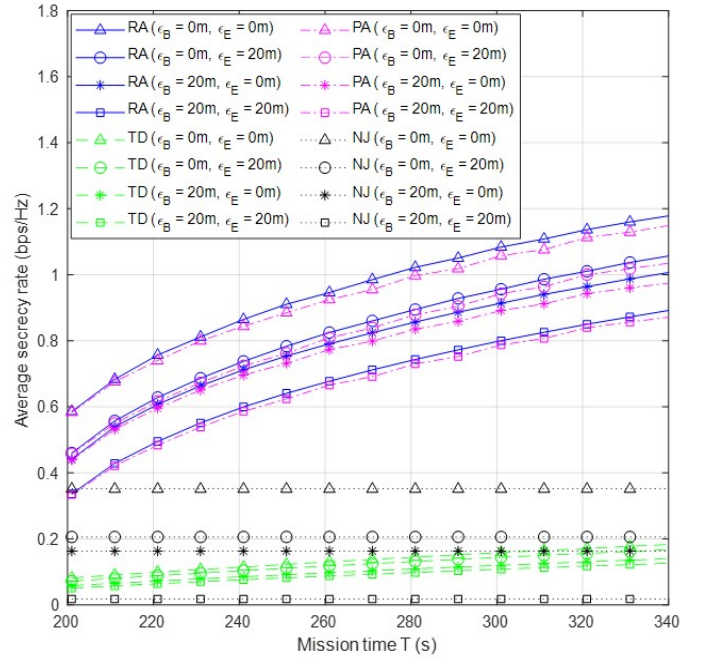


Fig. 5. Secrecy rate versus the mission time T

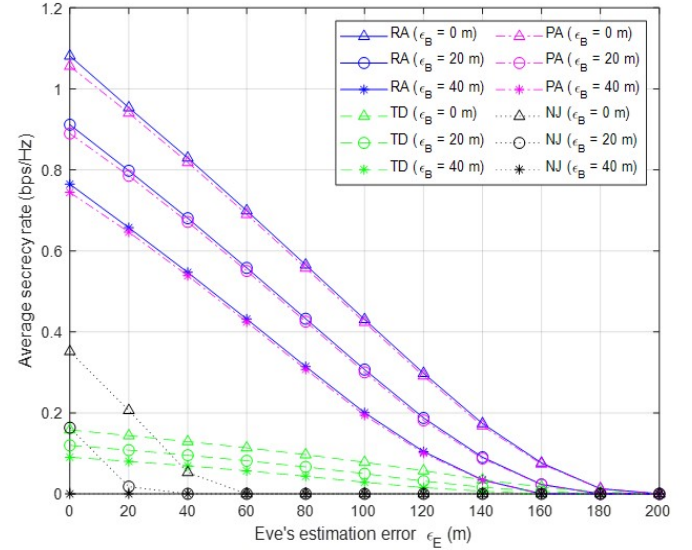


Fig. 6. Secrecy rate versus the estimation error ϵ_E , when the mission time $T = 300$ s

aforementioned methods and no jammer case (named as NJ) which is not operating the UAV jammer. It is observed that the average secrecy rates increase with the mission time T except for NJ, and RA outperforms the other methods. Furthermore, it is seemed that TD has the worst performance among the methods, even including NJ, except the case when $\epsilon_B > 20$ m and $\epsilon_E > 20$ m. The reason is that TD cannot control the interference in Bob.

Fig. 6 depicts the average secrecy rates with respect to Eve's estimation error ϵ_E , when $T = 300$ s. RA guarantees

the outstanding performance among all cases. Furthermore, in Fig. 5 and Fig. 6, the secrecy rates of all methods are degraded as ϵ_B and ϵ_E increase. In particular, when the UAV has more accurate information on the locations of the nodes, TD cannot guarantee the physical layer secrecy compared to NJ. Because the UAV in TD cannot handle the jamming power that affects Bob as described above. However, the increases in uncertainties of the locations (when the total amount of ϵ_B and ϵ_E exceeds 30 m) make the performance of TD better than that of NJ.

V. CONCLUSION

This work investigated how to improve the physical layer secrecy on the terrestrial wiretap channel with a potential Eve by using the cooperative UAV jammer. In particular, we took into account the location's uncertainty of Bob and Eve considering the practical situation such as a battlefield. We proposed the robust suboptimal algorithm to obtain the UAV jamming power and the trajectory for maximizing the secrecy rate. It was verified that reliable security performance can be achieved by the proposed algorithm via simulation results. In addition, it was concluded that the physical layer secrecy can be improved when the UAV has the sufficient mission time and knows more accurate information about the locations of the nodes. Other expanding scenarios to take into account air-to-ground fading channels or the uncertainty of other nodes (e.g., Alice) remain as interesting further work.

ACKNOWLEDGMENT

This work was supported by Institute for Information & communications Technology Promotion (IITP) grant funded by the Korea government (MSIT) (No.2018-0-00831, A Study on Physical Layer Security for Heterogeneous Wireless Network).

REFERENCES

- [1] H. Kang, J. Joung, J. Ahn, and J. Kang, "Secrecy-aware altitude optimization for quasi-static UAV base station without eavesdropper location information," *IEEE Commun. Lett.*, vol. 23, no. 5, pp. 851-854, May 2019.
- [2] S. Jeong, O. Simeone, and J. Kang, "Mobile edge computing via a UAV-mounted cloudlet: optimization of bit allocation and path planning," *IEEE Trans. Veh. Technol.*, vol. 67, no. 3, pp. 2049-2063, Mar. 2018.
- [3] S. Jeong, O. Simeone, and J. Kang, "Mobile cloud computing with a UAV-mounted cloudlet: optimal bit allocation for communication and computation," *IET Commun.*, vol. 11, no. 7, pp. 969-974, Nov. 2017.
- [4] Q. Wang, Z. Chen, H. Li, and S. Li, "Joint power and trajectory design for physical-layer secrecy in the UAV-aided mobile relaying system," *IEEE Access*, vol. 6, pp. 62849-62855, Oct. 2018.
- [5] M. Cui, G. Zhang, Q. Wu, and D. W. K. Ng, "Robust trajectory and transmit power design for secure UAV communications," *IEEE Trans. Veh. Technol.*, vol. 67, no. 9, pp. 9042-9046, Sep. 2018.
- [6] Y. Cai, Z. Wei, R. Li, D. W. K. Ng, and J. Yuan, "Energy-efficient resource allocation for secure UAV communication systems," *arXiv preprint arXiv:1901.09308*, 2019.
- [7] C. Zhong, J. Yao, and J. Xu, "Secure UAV communication with cooperative jamming and trajectory control," *IEEE Commun. Letters*, vol. 23, no. 2, pp. 286-289, Feb. 2019.
- [8] A. Li and W. Zhang, "Mobile jammer-aided secure UAV communications via trajectory design and power control," *China Commun.*, vol. 15, no. 8, pp. 141-151, Aug. 2018.

- [9] A. Li, Q. Wu, and R. Zhang, "UAV-enabled cooperative jamming for improving secrecy of ground wiretap channel," *IEEE Wireless Commun. Letters*, vol. 8, no. 1, pp. 181-184, Feb. 2019.
- [10] G. Zhang, Q. Wu, M. Cui, and R. Zhang, "Securing UAV communications via joint trajectory and power control," *IEEE Trans. Wireless Commun.*, vol. 18, no. 2, pp. 1376-1389, Feb. 2019.
- [11] C. Liu, T. Q. S. Quek, and J. Lee, "Secure UAV communication in the presence of active eavesdropper," *Int. Conf. Wireless Commun. WCSP*, Nanjing, 9th, pp. 1-6, 2017.
- [12] S. Gezici, M. R. Gholami, S. Bayram, and M. Jansson, "Jamming of wireless localization systems," *IEEE Trans. on Commun.*, vol. 64, no. 6, pp. 2660-2676, June 2016.
- [13] S. Bhamidipati and G. X. Gao, "Locating multiple GPS jammers using networked UAVs," *IEEE Int. of Things Journal*, vol. 6, no. 2, pp. 1816-1828, Apr. 2019.
- [14] I. S. Gradshteyn and I. M. Ryzhik, *Table of Integrals, Series and Products*, 7th edition. San Diego, CA: Academic, 2007.